Title: DAIMON: Digital Artifacts, Interrelational Mental Outcomes, and Narrative identity in Youth
Call: HORIZON-CL2-2026-01-TRANSFO-04; Single Stage.
Deadline: 23 September 2026

Meeting 30/04/2026
Layer 1: Leisure Time
Tools: (Digital tools, social media) LLM companion
Activity: sport, music, personal research, interrelational dialogues
Practice involved: …

Medium Layer:
Epistemic Friction Score (EFS) for each pair of [activity + digital tool involved]

Layer 2: Measurement
Correlation between EFS and:
1. Cognitive Autonomy (critical thinking, empathy, epistemic autonomy, metacognitive)
2. mental health (loneliness, …)
3. educational (task-oriented performance) (frictionless interaction and average performance > struggle and high performance)
    - dialog success.

Pitch Abstract
A 14-year-old spends three hours a day interacting with Character.AI. The chatbot never disagrees, never hesitates, never says "I don't know," never asks the user to wait. It responds instantly, mirrors emotions, and adapts to the user's preferences. The interaction is smooth and uninterrupted. The question is not whether this is engaging. It clearly is. The question is what happens the next day, when the same teenager is asked to focus in class, to struggle with a difficult concept, or to engage with a teacher who challenges their assumptions.
This situation is no longer exceptional. Youth-facing AI systems are being deployed at scale, shaping cognitive habits during formative years. These systems operate largely outside educational and regulatory frameworks, yet they establish interaction norms that strongly influence attention, learning strategies, and emotional regulation. The design choices made now will become the baseline expectations of an entire generation. Once these expectations stabilize, changing them will be difficult.
From an engineering perspective, the core issue is not content or model capability, but interaction architecture. Most AI systems for young users are optimized for engagement, retention, and user satisfaction. To achieve this, they minimize friction: no delays, no resistance, no breakdowns, no forced decisions. This design logic is effective in the short term, but it conflicts with how learning and cognitive development actually work. Sustained attention, critical thinking, and emotional resilience depend on exposure to effort, uncertainty, disagreement, and limits. When interaction is designed to remove these elements, cognitive effort is displaced rather than supported.
Yet, how do specific AI interaction patterns affect attention, learning, and autonomy in young users, and how can alternative designs support these capacities instead of eroding them? And, if AI

systems are becoming everyday learning environments for young people, what kind of minds are we engineering through their interaction design?
Every AI interaction embodies a theory of mind. When a chatbot mirrors a teenager's emotions perfectly, it assumes emotional growth comes from validation rather than challenge. When it answers instantly, it assumes knowledge comes from retrieval rather than effort. When it maintains uninterrupted interaction, it assumes attention is something to capture, not something to train. These are not neutral choices. They are pedagogical assumptions implemented in code. DAIMON's contribution is to make these assumptions explicit, testable, and improvable.
This is why philosophy and engineering must work together at the design level. Philosophy is used to clarify concepts such as agency, dependence, assistance, and manipulation in ways that can be translated into technical specifications. Engineers then implement alternative interaction architectures that embody different assumptions: interaction boundaries that force user choice, intentional delays, refusal modes, reduced emotional mirroring, and prompts that shift initiative back to the user. AI systems become experimental platforms where interaction parameters are systematically controlled. Psychology and mental health research measure the effects on attention, learning performance, and emotional regulation. The urgency of DAIMON lies in timing. Youth-facing AI systems are rapidly normalizing frictionless interaction as the default. If this becomes the standard before evidence-based alternatives are developed, future regulation will be limited to reactive restrictions rather than informed design requirements. DAIMON intervenes at the only point where meaningful control is still possible: before interaction patterns harden into norms. The project's outputs are designed for policy and methodologies impact. DAIMON will deliver empirical evidence linking specific interaction patterns to cognitive outcomes and translate these findings into design guidelines for youth-facing AI. These guidelines can inform standards, public procurement, and regulation by identifying which interaction architectures undermine cognitive autonomy and which support it. This enables ex ante governance focused on system behavior rather than post-hoc user responsibility or usage limits.

Project Draft

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Abstract

1. Context and Motivation

In response to HORIZON-CL2-2026-01-TRANSFO-04's call to investigate **digital tools used outside school for leisure and communication**, DAIMON addresses a critical paradigm shift: European youth are transitioning from passive media consumption to active AI-mediated relationships. As Generative AI chatbots, social robots, and AI companions enter the leisure sphere, these digital artifacts fundamentally reshape young people's **Epistemic Agency** (the capacity to focus, validate truth, and learn autonomously) and their **emotional resilience**.
Current policy responses focus on screen-time limitation and data privacy, but this is insufficient. DAIMON proposes a radical new framework based on **Epistemic Digital Autonomy**, the right to mental self-determination free from manipulative algorithmic design, often included in most of the digital tools design. We hypothesize that **'frictionless' interactions erode youths' tolerance for the productive friction required for education and empathy**, with measurable impacts on attention spans, critical thinking, and mental health indicators.

2. The DAIMON proposal: A Transdisciplinary Approach
Coordinated by a Philosophy of Technology unit (University of Padua and Pegaso University), DAIMON operationalizes deep conceptual analysis into rigorous empirical science. The consortium unites Psychology and Mental Health experts (Università Cattolica di Milano), AI researchers and developers (FBK Trento), NGO (Mind Care 4all), and Industrial partners (Mentalab), ensuring both academic rigor and relevance of the outcomes for the stakeholder**.**
The consortium is anchored by the coordinating leadership of the FISSPA Department at the University of Padua, which fosters a distinctively innovative synergy within the European academic landscape by bridging the disciplines of Philosophy, Sociology, Pedagogy, and Applied and Theoretical Psychology.

[Motivations underlying the composition of the consortium]

Main Hypothesis:

> *Our central hypothesis is that **frictionless interaction** with compliant AI agents erodes the youths' ability to tolerate the productive friction required for education, personal growth and epistemic autonomy.*

Reinforce the idea that frictionless designs are not good (literature).

The overarching objective of the project is to foster cognitive and epistemic autonomy through the informed use of the digital tools considered, facilitating a shift from frictionless interaction to critical engagement. The pivotal juncture of this transition lies in the critical discernment of what constitutes a conscious, self-determined usage, versus what conversely descends into manipulation and external direction or control.

Alignment with TRANSFO-04 Expected Outcomes:

- **Evidence Base for policymakers**: Deliver robust empirical data on how digital leisure tools affect learning capacity, attention span, and mental health.
- **Mechanisms of impact**: Identify specific design patterns (infinite scroll, variable rewards, anthropomorphic manipulation) that undermine educational outcomes.
- **Mechanisms of mitigation**: Identify specific software designs and methodologies that allow to select the degree of *frictional interaction* a system must have.
- **Policy recommendations**: Provide actionable guidelines for "Cognitive Ergonomics" labeling and regulation of youth-facing AI
- **Stakeholder engagement**: Co-design interventions with educators, parents, and youth themselves*** (tbd)

To this end, the project will implement a theoretical phase primarily oriented towards a fundamental empirical-practical development (WP1 & 2), followed by a technical-experimental phase designed to operationalize the hypotheses formulated in the initial stage (WP 3 & 4) [and a stakeholder-based WP5 tbd].

3. Methodology

The Conceptual Analysis (WP1):

We classify the digital tools involved not by content, but by "relational intensity" and by "design patterns". In other words, we analyse the digital tools not simply as mere tools but as **practice-shaping** artifacts:

- **Digital companion ontology**: Analyse the shift from "cognitive tool" to "cognitive partner." How do current Large Language Models (LLMs) and general-purpose-chatbot simulate or embody empathy, trustworthiness, and other fundamental properties of relational and epistemic interactions?
- **Epistemic Friction:** Their primary task would be to elaborate a rigorous definition of *epistemic friction*, understood as the cognitive and affective resistance encountered when a knowing subject engages with material, problems, or interlocutors. Drawing on epistemology and philosophy of education, we would distinguish epistemic friction from

mere difficulty or frustration, grounding it as a necessary condition for genuine understanding and belief formation. We would further differentiate *productive* from *unproductive* friction, mapping the conditions under which resistance promotes epistemic growth rather than disengagement. This conceptual scaffold would allow the project to operationalize "frictionlessness" as a measurable epistemic variable, identify which specific cognitive capacities AI compliance could erode, and normatively evaluate what forms of friction educational and technological design must preserve.

- **Definition and boundaries of Digital Cognitive Autonomy // *****: We will give an operative definition of the concept of cognitive autonomy, grounded in the academic debate, for empirical testing, and develop a scale that differentiates, encompassing intermediate states, between a condition of algorithmic passivity, where the human agent is fully hetero-directed, and one of conscious, critical engagement, where the human is fully cognitive autonomous and capable of epistemic autonomy and critical thinking.

Technical analysis and grounding for the Epistemic Friction Score (WP2):

- **Conceptual reverse-engineering and technical comparative analysis**: Preliminarily, rather than extracting source code from a finished product, we seek to reconstruct the conceptual matrix underlying the application. Every technological artifact embodies a specific theory, which is inevitably inscribed onto its outputs through direct or indirect transmission. From an educational perspective, our objective is to decode the dominant pedagogical, linguistic, and cognitive models governing AI models' generation behaviours and compare them to specific implemented technical solutions. This allows us to isolate and compare distinct models on two levels: a theoretical level (qualitative analysis) and a technical and performance-based level (quantitative analysis)
- **Developing the "Epistemic Friction score"**: We will elaborate a standardized way to measure how much "frictional interaction" a system realises. We hypothesize that non-productive AI requires near-zero cognitive effort and tends toward user compliance (frictionless design). Conversely, AI harnessed for critical formation demands effortful engagement. Such tools inevitably generate a higher cognitive load. Consequently, our guidelines will be based around different benchmarking methods developed to actively measure the full spectrum of the human-machine interaction. Specifically, the critical dimensions identified by the project (such as trustworthy vs. untrustworthy, relational vs. extractive, and effortful vs. frictionless) will not be treated as abstract categories, but as variables with precise calculability. The goal is to associate specific technical benchmarks with each of these oppositions: we will define metrics that integrate quantitative performance with computable qualitative assessment, thereby measuring the effective pedagogical adherence and safety of the models in real-world scenarios.
- **Analyse and Develop methodologies to influence the "Epistemic Friction score"**: We will analyse how the post-training data influence the behaviour of LLM-based system, assessing whether an instruction increases or decreases Epistemic Friction in the interactions. By analysing these mechanisms, we will be able to modify them, proposing concrete methodologies to increase or decrease epistemic friction in LLM-based systems, trying to assess with educational and psychological studies and trials the optimal level of epistemic friction based on parameters such as education, age, situation, system, application.

The "Dual-State" RCT (WP3-WP4; Divide into theoretical protocol and actual experiment): A controlled trial, involving 2000 students (chosen across different age groups, socio-economic backgrounds, cultural contexts, with or without disabilities), analyses the impact of LLM-based systems interaction on cognitive focus. Using a novel "Trust Trap" protocol, we measure **Epistemic Agency**: when an AI companion deliberately "hallucinates" a subtle error, does the student correct it, or do they surrender their critical judgment to the machine? [more…]
A distinctive element of our proposal lies in the **depth of the technical analysis** applied to the AI models powering today's chatbot, understood as digital companions. To address the opacity of proprietary systems, the consortium will focus on the **internal development of dedicated testing models** (probing frameworks). This approach allows us to go beyond superficial evaluation and technically operationalize the relational dynamics of digital tools, helping in formulate grounded metrics and guidelines, to deepen the preliminary conceptual understanding of the artifact's ontology and, furthermore, to adapt the post-training instructions of the models to achieve active strategies to reach optimal level of epistemic friction, in relation to the trial's results.

Stakeholder engagement (WP5): Co-design interventions with educators, parents, and youth themselves*** (tbd)

- Schools

4. Expected Outcomes and Policy Impact (deliverables) (1-3)
DAIMON will translate these findings into three concrete policy products, directly addressing the expected outcomes of the TRANSFO-04 call:

- **1. The "Epistemic Friction" Labeling System:** A consumer-facing framework that rates digital apps based on their cognitive load and epistemic friction. This provides parents and educators with actionable criteria for choosing "healthy" digital leisure tools and optimal or sub-optimal values of an **Epistemic Friction Score** based on parameters such as education, age, situation, system, application.
    - *Accomplished if*: found quantitative measurement valid for assessing the Epistemic Friction and creating an Epistemic Friction Score.
- **2. Creation of the "Epistemic Friction" Methodologies Dataset (Dare un Nome):** A dataset comprehensive of all methodologies to implement in post-training sessions to fine tune LLM-based software on the appropriate level on the **Epistemic Friction Score**.
    - *Accomplished if*: create a dataset of impactful instruction for post-training phase in order to modify the behaviour of a model, with the aim to influence the Epistemic Friction score.
- **3. The "Digital Cognitive Autonomy" Policy Framework:** A set of regulatory "Red Lines" for the design of AI companions targeting youth. We will propose specific bans on "frictionless algorithms" in educational contexts, advocating for **Constructive Friction** as a mandatory design principle for youth-facing AI.
    - *Accomplished if*: realise a policy framework compatible with current law on AI-system and further developments.
- **3b. (accorpare) Guidelines on "Digital Literacy" & Digital Cognitive Autonomy:** A curriculum module realised in par with the stakeholders that moves beyond basic digital skills to teach "critical reverse-engineering," empowering students to recognize how algorithms shape their emotions and attention.

- *Accomplished if*: realise published literature for public and stakeholder interest in order to ground the correct-use of AI-system in youth age in scientific literature.

Conclusion

DAIMON represents a pivotal shift in addressing the digital well-being of European youth, moving beyond traditional screen-time metrics to protect both Cognitive and Epistemic Autonomy, Epistemic Agency in an era of AI-mediated relationships. By leveraging a uniquely integrated consortium, … , the project transcends reactive policymaking.

Through its innovative investigation into frictionless design patterns and Trust Trap protocols, DAIMON will deliver the empirical evidence and Cognitive Ergonomics frameworks necessary to ensure that AI companions function as tools for extending critical thought rather than eroding it.

Working Team divided per unit:

Unit	**Referent**	**Organization***	**Location**	**Project Roles**
Conceptual - Ethical Analysis	Fabio Grigenti	FISPPA Department	Università degli Studi di Padova	Consortium Coordinator; WP1 - WP7 - WP8
Media Change and Innovation	Desirée Schmuck	Department of Communication	University of Vienna	
Technical Model Design	Bernardo Magnini	Natural Language Processing unit (NLP)	FBK - Fondazione Bruno Kessler (Trento)	WP3
Psychology	Antonella Marchetti - Davide Massaro	Psychology Department	Università Cattolica (Milano)	WP4
	Ioana R. Podina	Mind Care 4ALL	Bucarest	WP5
	Mireia Vendrell Morancho	Artificial Intelligence research Institute - Spanish National Research Council	Barcelona	WP4
Philosophy of Technology	Marco Tamborini	Dipartimento di Filosofia	Pegaso University	WP2
Schools and Sport centers				

Industrial Partner	Christoph Aurnhammer	Mentalab	Munich	Collaboration in WP3 - WP5

Tentative Timeline:

- End of April / Early May: Consortium 1st meeting.
- 15 May: call opening.
- End of June: first draft (each group works on its part) + meeting.
- Early July: Project is sent to Research Offices.
- *End July: Research Office draft (budget…)*
- 31st August: Final Draft.
- **Final Deadline: 23 September 2026**